TECHVAULTS

COOPERATIVE MANAGEMENT PLATFORM · STAGE 1 DISCOVERY

Cooperative Sector Discovery and Research Plan

How TechVaults will learn how Nigerian cooperative societies actually operate, and how those findings become a configurable product.

VERSION
0.2

DATE
17 October 2026

STATUS
Plan accepted · fieldwork not started · platform foundation in preview

AUDIENCE
TechVaults stakeholders

Understand how cooperatives work before building software to manage them.

Nothing in this document is legal, accounting or regulatory advice. Named experts and organisations are learning references only and have not endorsed TechVaults. Sources were checked on 17 October 2026.

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EVIDENCE LABELS

VERIFIED confirmed in a primary or official source

REPORTED press, commentary or vendor claim, not independently confirmed

HYPOTHESIS our working belief, to be tested

RECOMMENDATION proposed course of action

OPEN unanswered question

Overview and progress report

TechVaults Cooperative Management Platform · Stage 1: Discovery Version 0.1 · Prepared 17 October 2026 · Status: Draft for internal review

> Guiding principle: understand how cooperatives work before building software to manage them.

This folder is the first deliverable from the master prompt (§16). It is a **plan and a set of research instruments**. It does not specify the product. No code, schema or stack decisions should be taken from it.

Evidence labels used throughout

Each claim carries one of these labels, so readers can see how far to trust it.

Label	Meaning
VERIFIED	Confirmed in an official or primary source, which is cited. Still subject to legal review before we build anything on it.
REPORTED	Stated by credible secondary sources (press, law-firm commentary, vendor sites). Not independently confirmed.
STAKEHOLDER	Opinion or practice reported by an interviewee. Applies to their society only. *(No interviews yet, so this label is not used in v0.1.)*
HYPOTHESIS	Our working belief. It needs testing.
RECOMMENDATION	Our proposed course of action.
OPEN	An unanswered question.

Progress report (format from master prompt §15)

Current understanding - VERIFIED The Federal Ministry of Agriculture and Food Security (FMAFS) calls itself the supervising ministry for cooperative development. The Nigerian Co-operative Societies Act (Cap N98 LFN 2004) is under review, and the plan is to align it with the ICA-Africa Model Cooperative Law (FMINO, 30 Jan 2026). - VERIFIED On 12 August 2026, FMAFS signed an MoU with Seamfix for a National Cooperative Smart Registry (NCSR). The NCSR issues a Cooperative Verification Number (CVN) to each society and a CoopID to each member. A six-month revalidation of all cooperatives, run through State Departments of Cooperatives, started at the same time (FMINO, 12 Aug 2026). - VERIFIED The NDPA 2023 is now implemented through the NDPC's GAID 2025, which took effect on 19 September 2025 and replaced the NDPR 2019 (DLA Piper, Afriwise). We still need to read the official text. - VERIFIED The text of Cap N98 was read from two independent copies. It provides a 25% statutory reserve (s.34(2)), an education fund capped at 10% (s.34(3)), a one-fifth share-holding cap (s.27), no loans to non-members (s.30), the society's charge and set-off over member shares and deposits (s.16), nominee-first transfer on death (s.18), year-end audit (s.36), and State Directors of Cooperatives (s.1(2)). The Act says nothing about payroll deductions. See 02 F17–F27. - OPEN We do not yet know how federal and state cooperative laws divide authority. We also do not know whether a

new Cooperative Societies Bill has been introduced, how the revalidation is going in each state, or whether the NCSR will offer integration APIs. - OPEN We have done no field interviews. Everything we know about how societies operate is still hypothesis.

Business impact. The federal digitalisation programme and the revalidation deadline (about February 2027, if six months from 12 August 2026 holds) give societies a reason to move off paper and spreadsheets now. That is a time-limited opening. It also carries a risk: the regulator may impose a data or identity standard we have not yet seen.

Recommendation (accepted 17 Oct 2026, see decision log). Run the 12-week discovery plan (08). Start in parallel with (a) desk verification of the legal questions, with a Nigerian cooperative lawyer, and (b) recruiting 8 to 12 societies, weighted towards staff (salary-deduction) cooperatives.

Implementation. Documents only. No code. Week 0 desk work is done: Act provisions extracted, competitor desk scan (06 §6.4), fieldwork kit drafted.

Verification. Every source was checked as of 17 October 2026. Each claim's label shows how far it was verified.

Open questions and risks. See 07 Assumptions register and the expert-review list in 02 §2.6.

Discovery Decision Log

These decisions were accepted by TechVaults on **17 October 2026** ("proceed with every recommended professional decision"). Each one can be revisited when evidence contradicts it. Record the reversal here; don't delete the entry.

ID	Decision	Rationale	Revisit when
D-001	**Research before build.** No product code, schema or stack lock-in until the Discovery Report (week 12) is accepted.	Master prompt §16	Discovery Report accepted
D-002	Run the **12-week discovery** starting **2 Nov 2026**, ending about **29 Jan 2027** (08).	Finishes before the NCSR revalidation window closes (about Feb 2027)	Recruitment falls behind by more than 2 weeks
D-003	Sample **8–12 societies, weighted to workplace (payroll) cooperatives**, with at least one community or agricultural and one multipurpose society, across **Lagos, the FCT and one other state** (01 §1.4).	Likely MVP segment; avoid a payroll-only core by accident	Fewer than 6 workplace societies recruited by week 4
D-004	Retain three advisers in weeks 1–2: **cooperative and financial-services lawyer**, **ICAN-qualified cooperative auditor**, **NDPA data protection officer or licensed compliance organisation**. Terms of reference in kit/.	Several requirements are blocked on their answers (02 §2.6)	n/a
D-005	**First pilot profile:** a mid-sized (300–2,000 members) single-employer **private-company or university staff cooperative** with savings and 2–4 loan products. **Exclude IPPIS-based federal MDA societies** from the first pilot (10).	Structured monthly cycle; IPPIS coordination issues reported (02 F27); bank staff coops are proven buyers (06 §6.4)	Discovery evidence shows a better-fitting segment
D-006	Treat the following as **non-configurable guardrails** pending counsel confirmation: 25% statutory reserve with Director exemption only (s.34(2)); education fund ≤10% (s.34(3)); 1/5 share-holding cap (s.27); no loans to non-members (s.30); nominee-first death settlement (s.18).	Statute text verified (02 F2, F17–F25)	Counsel memo or state law says otherwise
D-007	Configurability uses **parameters and named, accountant-verified strategies**, not a free-form formula language (09 §9.2).	Testable, explainable to auditors	A real society needs a rule no strategy can express
D-008	The platform is a **record-keeping and administration system**. It does **not hold or move funds or take deposits**. Any money movement goes through CBN-licensed providers, after legal review.	CBN deposit-taking restrictions (02 F15); Act s.32	Counsel opinion received
D-009	**Data minimisation:** no NIN, BVN or bank account numbers are collected during research. In the product, NIN or CoopID storage is optional and DPIA-justified.	NDPA 2023 / GAID 2025	DPO memo received
D-010	**UI direction (provisional):** shadcn/ui + Radix + Tailwind; TanStack Table and Query; AG Grid only for spreadsheet-grade reconciliation; React Hook Form + Zod; Storybook (11). Prototypes in weeks 10–11 use it.	Owned components, accessibility, premium feel	Stage 4 stack decision

ID	Decision	Rationale	Revisit when
D-011	Discovery documents live in this repository (`docs/discovery/`) under version control. A published read-only page is shared with stakeholders.	Single source of truth, change history	n/a
D-012	**Build starts in parallel with discovery (17 Oct 2026; partly supersedes D-001).** At TechVaults' direction, engineering starts on the **policy-independent foundation**: project scaffold, design system, authentication and MFA, multi-society tenancy and isolation, roles and permissions, append-only audit log, double-entry ledger core, and a basic member registry. **Policy-dependent modules** (loan rules, deduction allocation order, exit settlement, dividends) are still gated on discovery evidence and expert memos. They are built only as configurable strategies once the rules are validated.	The foundation is needed whatever the findings. Building policy logic before validation risks encoding the wrong rules (master prompt §§6–7)	Discovery findings contradict a foundation assumption
D-013	**Stack (supersedes the "provisional" status of D-010):** TypeScript end to end; **Next.js (App Router)** web app; **PostgreSQL** with row-level security for tenant isolation; **Drizzle ORM** and SQL migrations; money stored as **integer kobo (bigint)** with documented rounding; **Vitest** for domain tests and **Playwright** for end-to-end tests; pnpm. One deployable modular monolith, no microservices.	Widely known and maintainable; Postgres gives transactional integrity and RLS; integer minor units avoid float errors	Team capability or hosting constraints say otherwise

1. Cooperative Models to Study in Nigeria

The rows below are HYPOTHESIS. They are working descriptions built from the TechVaults brief and general sector knowledge. Interviews must confirm or correct each cell. Do not cite this table as fact.

1.1 Segment map

#	Segment	Typical membership	How money comes in	Likely distinguishing needs	Discovery priority
A	**University and educational staff** (federal, state, private universities, polytechnics, colleges, school staff)	Academic and non-academic staff, often a few hundred to several thousand	Monthly payroll deduction by the bursary	Several loan products (regular, emergency, car, housing). Staff transfers, sabbaticals and retirement. Possible effects of IPPIS or federal payroll.	**High**
B	**Government ministries, departments and agencies**	Civil servants. Registers can be large.	Payroll deduction, possibly through IPPIS or state payroll platforms	Formal approval hierarchies, audit scrutiny, posting and redeployment between MDAs, mass retirements	**High**
C	**Hospitals and healthcare** (teaching hospitals, federal medical centres, private hospitals)	Clinical and non-clinical staff, shift-based	Payroll deduction. Mixed employers (locums, consultants).	High staff turnover, welfare and bereavement schemes, reconciling against several payroll sources	**Medium–High**
D	**Private companies and corporate workplaces** (banks, oil and gas, manufacturing, telcos, NFIU-style agency coops)	Employees, sometimes including contract staff	Payroll deduction. The employer may also contribute.	HR-system file formats, recovering loans when someone leaves, confidentiality of salary data, employer as a stakeholder	**High**
E	**Professional associations** (NBA branches, NMA, ICAN districts, NUT)	Members spread across many employers	Bank transfer and direct debit. Very little payroll.	No single payroll source, so reconciliation is by bank statement. Members across states.	Medium
F	**Traders, artisans, market and community cooperatives**	Market associations, artisan guilds, community groups	Cash, agent collection, bank transfer, USSD. Often irregular.	Rotating savings (*ajo*/*esusu*), group lending and group guarantees, low-literacy and feature-phone users, cash handling controls	Medium (study now, build later)
G	**Farmers and agricultural cooperatives**	Smallholder farmer groups	Seasonal, harvest-linked, input loans in kind	Seasonal repayment, inputs, links to government programmes and NCSR farmer registration	Medium (study now, build later)

#	Segment	Typical membership	How money comes in	Likely distinguishing needs	Discovery priority
H	**Large multipurpose societies** (mixed membership, several branches)	Thousands of members, branches	Mixed	Many savings schemes, share capital and dividends, investments, consumer or commodity sales, branch accounting, more formal audits	Medium

The Federal Government's February 2026 plan to digitally register about 30,000 farmers' cooperatives (Nigeria Startup Act) suggests segment G may meet government digital systems sooner than other segments. REPORTED

1.2 Cross-cutting dimensions to record for every society

Classify every interviewed society on these axes, not just its segment label. Our configurable product will vary along the axes, not the labels.

1. **Collection channel:** payroll deduction, bank transfer, cash, agent, USSD or mobile money, or a mix.
2. **Employer relationship:** none, employer facilitates deductions, employer contributes, or employer guarantees.
3. **Legal registration:** state-registered, federally registered, or unregistered. Also NCSR revalidation status (CVN issued?).
4. **Size:** members, branches, annual turnover, loan book.
5. **Product breadth:** savings only, savings and loans, multipurpose (trading, investments, welfare).
6. **Share capital model:** fixed shares, variable shares, none.
7. **Surplus model:** dividends on shares, patronage refund, interest rebate, none.
8. **Governance formality:** staffed secretariat and external auditor, or volunteer officers.
9. **Current tooling:** paper, Excel, WhatsApp, accounting package, cooperative software.
10. **Digital literacy and device access** among officers and members.

1.3 Hypothesised commonalities (to test)

- HYPOTHESIS Segments A–D share one operating core: a monthly deduction cycle, a member ledger, loans backed by guarantors and savings multiples, and settlement on exit. If so, they are one product with configuration, not four products.
- HYPOTHESIS Segments F and G differ most. They involve irregular cash collection, group liability and field agents. They probably need a different collection module and a different mobile experience, even if the ledger core is shared.
- HYPOTHESIS Segment H is mostly A–D with branch accounting, investments and trading modules added.

1.4 Recommended discovery sample (8–12 societies)

Segment	Target interviews	Rationale
A University	2–3	Likely early market, many loan products, retirement-heavy

Segment	Target interviews	Rationale
B Ministry or agency	2	Payroll-platform complexity, large registers
C Hospital	1–2	Turnover and multi-source payroll stress-tests reconciliation
D Corporate	2	HR-system integration, likely ability to pay
E/F/G/H	2–3 (at least one of F or G, one H)	Prevent the core design from becoming payroll-only by accident

Aim for spread across at least **three states**, including Lagos and the FCT, plus one other state. State laws and supervisory practice may differ (02).

2. Regulatory Landscape and Research Sources

All sources were checked on **17 October 2026**. Nothing here is legal advice. Every item marked for expert review must go to a Nigerian lawyer with cooperative-law practice before we write any rule into the product.

2.1 Verified and reported facts

#	Finding	Label	Source
F1	The principal federal statute is the **Nigerian Co-operative Societies Act, Cap N98, LFN 2004**.	VERIFIED (it exists); text to be confirmed against an official consolidation	FMINO, 30 Jan 2026; FAOLEX copy
F2	**s.34:** no dividend, bonus or other distribution of net surplus except as the Act provides (s.34(1)). **At least one fourth of net profits**, as shown in the audit report, must go to a **reserve fund**. The Director may exempt a limited-liability society or reduce the rate (s.34(2)). With the Director's approval, **up to 10% of the remainder** may go to an **education fund** (s.34(3)). Wilful non-compliance is an offence (s.34(4)). An **unlimited-liability** society needs the Minister's or Commissioner's approval before any distribution (s.34(5)).	VERIFIED against two independent copies of the text (official gazette still to obtain)	FAOLEX copy, Cap N98 (1993 No. 90); Jet Lawyer copy; NFIU coop bye-law
F3	The **Federal Ministry of Agriculture and Food Security (FMAFS)** calls itself "the supervising Ministry responsible for cooperative development". The **Federal Department of Cooperatives** sits within it.	VERIFIED	FMINO, 30 Jan 2026
F4	In January 2026 FMAFS began reviewing Cap N98. It plans a harmonised document aligned with the **ICA-Africa Model Cooperative Law** and the **Africa Ministerial Declaration** (14th AMCCO, Kenya, October 2025). The review covers digital registration, a national database, digital membership ID and electronic filing and reporting.	VERIFIED (announcement), **not enacted law**	FMINO, 30 Jan 2026; Premium Times
F5	Reports say the ministry wants to **repeal and re-enact** the Act, working with the Nigerian Law Reform Commission, the Federal Ministry of Justice and the National Assembly.	REPORTED	Guardian; TG News
F6	Earlier **amendment bills** (HB 71 of 2019; 2023 and 2024 bills on stiffer penalties) appear in National Assembly and NILDS records. We found no evidence that any of them passed.	REPORTED; status OPEN	NASS Bill Tracker; NILDS 2023; NILDS 2024
F7	On **12 Aug 2026**, FMAFS signed an MoU with **Seamfix** to create the **National Cooperative Smart Registry (NCSR)**. Revalidated societies receive a **Cooperative Verification Number (CVN)**. Members receive a **CoopID** over time, linked to the national identity infrastructure. Some reports also name the **Cooperative Federation of Nigeria (CFN)** as a party.	VERIFIED	FMINO, 12 Aug 2026; Nairametrics
F8	A **six-month national revalidation** began at the same time. Societies must have their registration records verified by their **State Department of Cooperatives**. State Directors are called "frontline regulators and implementers".	VERIFIED. End date not published; about Feb 2027 by inference.	Same as F7; FMINO: states charged

#	Finding	Label	Source
F9	Reports say CoopID links to the **NIN**, and USSD access is planned.	REPORTED	Biometric Update; ID Tech
F10	Sources disagree on the number of registered cooperatives. The January release says "**over 30,000**". August reports say "**more than 370,000**".	CONFLICT, OPEN	F3 and F7 sources
F11	In **Lagos**, the Ministry of Commerce, Industry and Cooperatives supervises societies through its **Department of Cooperative Services**. Lagos has its own **Co-operative Societies Law**.	REPORTED; obtain the law text	Lagos MCCTI; Nairametrics 2021; LawNigeria
F12	Practitioner guides say registration happens at the state cooperative ministry. They quote minimum membership of **20** (savings and credit), **50** (agricultural) and **10** (other).	REPORTED (secondary)	LawCare Nigeria; Mondaq
F13	Stakeholders have proposed a **Cooperative Development Commission**. It is a proposal only.	REPORTED	NAN, Jul 2026
F14	**NDPA 2023** is in force. The NDPC's **GAID 2025** was issued 20 Mar 2025, took effect **19 Sep 2025** and replaced the NDPR 2019. It covers compliance audit returns, DPIAs, DPO credentials and breach reporting.	VERIFIED via law-firm commentary; read the official text	DLA Piper; Afriwise; Mondaq
F15	The **CBN** says only deposit money banks, microfinance banks and primary mortgage institutions may take deposits **from the public**. We found no CBN rule that names cooperatives specifically.	REPORTED; scope for cooperatives OPEN	CBN MFB guidelines
F16	**ICA principles:** voluntary membership, democratic member control (one member, one vote in primary societies), member economic participation (reserves, part of them indivisible, and allocation in proportion to transactions), autonomy, education, cooperation among cooperatives, and concern for community.	VERIFIED	ICA cooperative identity

Provisions of Cap N98 (added 17 Oct 2026)

Read from the FAOLEX copy (Decree 1993 No. 90, commenced 26 Aug 1993) and cross-checked with the Jet Lawyer copy. VERIFIED: TEXT. What the text means, and how it applies alongside state laws, still needs counsel.

#	Provision	Product implication HYPOTHESIS
F17	**s.1:** the President appoints a Federal Director of Co-operatives (s.1(1)). **A State Governor may appoint a State Director of Co-operatives** (s.1(2)).	The Act itself provides for directors at both federal and state level. The system must record which Director (registrar) a society answers to.
F18	**s.3(1):** a primary society needs **at least ten persons**. s.3(2): an industrial society needs at least six.	This conflicts with the "20 for savings and credit" in practitioner guides (F12). That figure may come from state law or regulations. **OPEN.**
F19	**s.27:** no member, other than a registered society, may hold **more than one fifth of share capital**.	A share-capital cap (20%) that the system enforces by default.

#	Provision	Product implication HYPOTHESIS
F20	**s.16:** the society has a **charge on the shares, interests and deposits** of present, past and deceased members for debts owed to it, and may **set off** amounts owed against dividends, bonuses or profits payable.	Supports netting loans against shares and deposits at exit. The exact order and scope should come from bye-laws plus counsel.
F21	**s.18:** on a member's death, their interest goes to the **nominee**. If there is none, it goes to the **heir or legal representative**. The society may instead pay out the value.	The member record needs a nominee designation, kept separate from next of kin. Death settlement has a statutory order of recipients.
F22	**s.30:** the society may **not lend to non-members**, except another registered society with majority member consent. **s.32:** other dealings with non-members are subject to prescribed restrictions.	Loans may only go to active members. Exited members with residual loans need a defined treatment.
F23	**s.36:** accounts are **audited at each financial year-end** by an auditor **approved in writing**. The Director may prescribe returns (s.36(7)) and has access to all books (s.36(6)).	Year-end close workflow, an auditor role with read access, and a returns generator driven by Director directions (state-specific).
F24	**s.49:** certain disputes go **to the Director** for settlement or arbitration. Appeal to the Minister or Commissioner within 30 days.	A dispute register with an escalation path.
F25	**s.11:** bye-laws must cover member withdrawal and expulsion and any payment due to them, and member loans (interest, maximum amounts, repayment periods).	Confirms that **exit payouts and loan terms are bye-law-defined, so they are configurable policy**, within statutory guardrails.
F26	The Act contains **no provision on salary or payroll deductions**.	Payroll deduction rests on employer arrangements and member authorisation, not the Act. Collect the deduction authority (D05).
F27	**IPPIS** (OAGF) handles third-party remittances, and cooperatives are one listed category. Reports describe the HR role player creating the cooperative on IPPIS before deductions can start. Academic work reports **poor coordination** between cooperatives and the IPPIS office on recovering loans.	**REPORTED.** Supports keeping federal-payroll societies out of the first pilot. Get the official procedure from OAGF. Sources: Guardian, ResearchGate study

Note on age: the Act's penalty figures (e.g. s.34(4)) date from 1993, which is one reason a repeal is under way (F4–F5). Do not build penalty logic into the product.

Product implications (all HYPOTHESIS until legal review)

- **Reserve-fund rule (F2):** dividend and surplus logic needs a statutory minimum reserve (25%) that the platform enforces, with a recorded Director exemption as the only override. Societies should not be able to configure it away. Also needed: an optional education-fund appropriation capped at 10%, and a Minister or Commissioner approval step for unlimited-liability societies.
- **NCSR (F7–F9):** every society record should store its CVN. Member records should store a CoopID field. NIN storage should be optional and justified under NDPA. A future NCSR integration should be designed for but not built until an API is published.
- **CBN (F15):** the platform must stay a **record-keeping and administration tool**. It must not take deposits from non-members, and money movement must go through licensed providers.

2.2 Regulators and authorities to investigate

Body	Why it matters	What to obtain
Federal Ministry of Agriculture and Food Security, Federal Department of Cooperatives	National policy, Act review, NCSR	Current Act text, draft bill, NCSR data standards, revalidation guidelines
State Departments or Ministries of Cooperatives (Lagos, FCT, plus at least one other state, e.g. Oyo, Kaduna, Enugu or Rivers)	Registration, supervision, audit, revalidation	State cooperative law and regulations, model bye-laws, audit and annual return forms, filing calendar
Cooperative Federation of Nigeria (CFN) and state federations	Apex body, NCSR partner	Member surveys, common practices, introductions to societies
Nigeria Data Protection Commission (NDPC)	NDPA and GAID compliance	Official GAID text, DPIA template, audit-return thresholds
Central Bank of Nigeria (CBN)	Boundary of deposit-taking, payment providers	Applicable circulars; position on cooperatives and fintech partners
NIMC	NIN verification if used	Terms for verification-service use
Office of the Accountant-General / IPPIS (federal), state payroll bureaux	Third-party deductions for public servants	Deduction-schedule formats, remittance timelines, onboarding rules for cooperatives as deduction beneficiaries
National Assembly, NILDS, Nigerian Law Reform Commission	Bill status	Bill text and stage
FIRS and state IRS	Tax treatment of cooperatives and dividends	Exemptions; withholding on dividends or interest (expert review)
Financial Reporting Council of Nigeria (FRC)	Reporting framework applicability	Whether IFRS, IFRS for SMEs or another framework applies to cooperatives
SCUML / NFIU	AML obligations, if any apply to cooperatives	Registration requirements (expert review)

2.3 Sector organisations and standards

- **International Cooperative Alliance (ICA)** and **ICA-Africa**: principles, Model Cooperative Law (the stated reference for Nigeria's reform). ica.coop
- **ILO**: Recommendation 193 (Promotion of Cooperatives), cooperative statistics guidelines.
- **WOCCU**: PEARLS financial monitoring ratios, credit-union governance and the *Model Law for Credit Unions*. Useful for defining reports and risk indicators. woccu.org
- **ICAN, ACCA, IFRS Foundation**: accounting treatment of member shares (IFRIC 2 *Members' Shares in Co-operative Entities* is directly relevant), loan impairment (IFRS 9 or simplified approaches) and SME reporting.
- **OWASP (ASVS, Top 10), CIS Controls**: security baseline.

2.4 Research sources to mine

- Nigerian academic literature: cooperative economics departments (UNN, Unizik, OAU, ABU), the *Nigerian Journal of Cooperative Economics and Management*, and the Federal Cooperative College (Ibadan, Kaduna, Oji River).
- NILDS working papers on the amendment bills.
- Published society bye-laws, e.g. the NFIU Cooperative bye-law, as a starting corpus for policy variations.
- Press: Nairametrics, Premium Times, BusinessDay, The Cooperator News, for regulatory tracking.
- Vendor sites, as evidence of market positioning only (06).
- Open-source reference: **Apache Fineract / Mifos X**, for loan-product and ledger modelling. Study it; do not assume it fits Nigeria.

2.5 Federal vs state: how to resolve the question

We could not confirm from primary sources how the Constitution divides cooperative-law powers between federal and state governments, or how far the federal Act binds state-registered societies. OPEN The Act itself provides for State Directors (F17), and states such as Lagos have their own laws (F11). That suggests both layers operate. How they rank against each other is a question for counsel. Steps: 1. Get the current text of Cap N98 and of state laws for Lagos, the FCT and one other state. 2. Ask counsel how the Act and state laws relate, and which law governs a society registered in state X whose employer is federal. 3. Record the answer as a jurisdiction matrix covering registration, audit, annual returns, reserve rules, dispute resolution and liquidation.

2.6 Questions requiring expert review

Legal (Nigerian cooperative and financial-services counsel) 1. Which law governs a society: the federal Act or the state law? How do they interact? 2. Statutory minimum reserve, distribution approvals, audit requirements (who may audit, how often, filing deadline). 3. Is taking savings from members only clearly outside CBN licensing? What about non-member deposits or online collection? 4. Can loan interest be charged freely, and is there any usury or disclosure requirement? 5. Settlement on death: what must be paid to the nominee or next of kin, and what needs probate or letters of administration? 6. Legal status of electronic records, e-signatures and electronic AGM votes under current law (Evidence Act 2011 s.84; Cybercrimes Act). 7. Required record-retention periods.

Accounting (ICAN/ACCA-qualified cooperative auditor) 8. Are members' shares equity or a liability (IFRIC 2)? Should withdrawable savings be shown as liabilities? 9. Interest recognition methods (flat, reducing balance, upfront deduction) and how to present them. 10. Loan impairment and provisioning approach expected by state auditors. 11. Chart of accounts standard used by state audit departments. 12. Dividend and patronage-refund computation and period cut-off.

Data protection and security (NDPC-registered DPO or compliance organisation) 13. Are TechVaults and each society data controllers, processors or joint controllers? 14. Is a DPIA required (financial data at scale, NIN)? Do we fall in the GAID audit-return tiers? 15. Is data localisation required, and what are the cross-border hosting rules? 16. Lawful basis for processing next-of-kin and guarantor data.

3. Stakeholder Interview Guides

3.1 How to run interviews

- **Format:** 45–60 minutes, ideally in person and where the work happens (the treasurer's office, the bursary). Two people from TechVaults: one asks, one takes notes.
- **Consent first:** read the consent script (§3.11). Record audio only with explicit consent. Never photograph a document showing member names, account numbers, salaries or NINs. Ask for an anonymised copy instead (05).
- **Ask about the past, not the hypothetical** (Teresa Torres, *Continuous Discovery Habits*). "Tell me about the last time a member retired" beats "How would you like retirement to work?"
- **Ask them to show you.** "Can you show me the spreadsheet you used last month?" Watch for workarounds, copying between files, and calculators.
- **Don't pitch.** Don't show a product or describe features until the final 5 minutes, if at all.
- **Log each answer with its source.** Every finding is `STAKEHOLDER: SOCIETY ID, ROLE`. One society's practice is never "how cooperatives work".
- **Same-day synthesis:** fill in the interview snapshot (§3.12) within 24 hours.

3.2 Opening questions (all roles, 5 minutes)

1. What is your role in the society, how long have you held it, and is it paid or voluntary?
2. Roughly how many members does the society have, and how has that changed?
3. Walk me through what you did for the society last week.
4. Which part of your cooperative work takes the most time? Which part worries you most?

3.3 Chairperson, executive committee, principal officer

1. How are decisions made: committee meetings, AGM, delegated officers? How often do you meet?
2. Which decisions need committee approval, and which need AGM approval? Are there value thresholds?
3. Tell me about the last bye-law or policy change. How was it proposed, approved and communicated?
4. How do you know the society's finances are healthy? What reports do you look at, and how often?
5. What do the State Department of Cooperatives or your auditors ask you for, and when?
6. Have you started the national revalidation (NCSR, CVN)? What has the experience been like?
7. What complaints do members bring to you most often?
8. Describe the last dispute involving a member, a loan or a guarantor. How was it resolved?
9. How are elections held, and how are signatories changed afterwards?
10. Has the society ever discovered fraud or a serious error? How was it found?
11. Who chose your current tools (books, Excel, software)? What would make you change them? Who would decide, and with what budget?

3.4 Secretary or administrator

1. Walk me through how a new member joins, from first enquiry to their first deduction.
2. What forms and documents do you collect? Can we see a blank copy?

3. How are member numbers assigned? What happens to the number when someone leaves and rejoins?
4. How do you find out that a member has changed department, been transferred, retired or died?
5. How do you keep records up to date (address, next of kin, bank account)? Who is allowed to change them?
6. Have you ever found a duplicate member, or someone deducted who wasn't a member?
7. How do you communicate with members: notices, WhatsApp, SMS, email?
8. How are minutes, resolutions and correspondence stored?
9. What files would you grab first in a fire?

3.5 Treasurer or accountant

1. Show me how you close a month. What do you start with and what do you produce?
2. What accounting records do you keep: cashbook, ledgers, journals, trial balance? In what tool?
3. How many bank accounts does the society have, and who can sign?
4. When the payroll remittance arrives, how do you match it to individual members? How long does that take?
5. What happens when the remittance is short, late, or includes people who aren't members?
6. How do you split each member's deduction between savings, shares, loan principal, interest and other items? Who decides the order?
7. How do you calculate loan interest? Show me an example of a real (anonymised) loan.
8. How do you produce a member's statement? How often do members ask for one?
9. Describe your bank reconciliation process. How often is it done, and by whom?
10. How do you correct a mistake once it has been entered?
11. How are year-end accounts prepared, who audits them, and what do auditors usually query?
12. How are dividends or surplus computed and paid? Can we see last year's computation?
13. What investments, fixed assets or trading activities does the society have?
14. What are the payment approval steps: vouchers, signatories, limits?

3.6 Loan officer or loan committee member

1. What loan products do you offer? For each: maximum amount, term, interest rate and method, fees, eligibility.
2. Walk me through the last loan approved, from application to disbursement. How many days did it take?
3. How do you check eligibility (months of membership, savings multiple, salary capacity, existing loans)?
4. How do guarantors work? How many are required, what must they have, can a person guarantee several loans, and what happens to the guarantor if the borrower defaults?
5. Who approves which loans? Are there limits by amount or by who the applicant is (for example, a committee member)?
6. How do you check that the employer's payroll can still take the deduction (the net-pay floor)?
7. How is a loan disbursed: cheque, transfer, offset against another loan?
8. What happens with top-ups, refinancing, rescheduling and early repayment?
9. How do you track arrears? What is done at 1, 3 and 6 months overdue?
10. Tell me about the last loan written off or recovered from guarantors or savings.
11. How do you handle loans to someone due to retire within the loan term?
12. Are there queues or rationing because funds run short? How are applicants prioritised?

3.7 Payroll or HR officer (employer side)

1. How does the cooperative's deduction get onto the payroll? Who sends what, in what format, and by what date?
2. Which payroll system do you use (IPPIS, state platform, SAP, Sage, a bespoke HR system)? Can we see a blank deduction template?
3. How do you handle a deduction that exceeds the member's net pay or a statutory cap?
4. How and when is money remitted to the cooperative? Is there a remittance schedule?
5. What do you send back to the cooperative: a deduction report, exceptions?
6. How do you notify the cooperative of resignation, retirement, transfer or death?
7. What problems come up between payroll and the cooperative?
8. What data-sharing agreements or approvals exist between employer and cooperative?

3.8 Internal or external auditor

1. Which framework and standards do you audit cooperatives against?
2. What records do you ask for? Which are most often missing or inconsistent?
3. How do you test member balances, the loan portfolio and interest income?
4. What are your most common findings and management-letter points?
5. What do you need from a software system to rely on it (audit trail, change logs, read-only access, exports)?
6. What do you file with the State Department, and when?
7. How do you treat member shares, savings, loan impairment and reserves?

3.9 Members (including recently retired or exiting members)

1. Why did you join? What do you use the cooperative for?
2. How do you know how much you have saved and how much you owe? When did you last check?
3. Tell me about the last time you applied for a loan. What was hard? How long did it take?
4. Have you ever been a guarantor? How did that go?
5. Have you ever disagreed with your balance or a deduction? What happened?
6. *(Retired or exited)* Walk me through leaving the society. How long did settlement take? Were you satisfied with the calculation?
7. What phone do you use? Do you use banking apps, USSD, WhatsApp?
8. Who do you trust in the society, and why?

3.10 Closing (all roles)

1. If you could fix one thing about how the society runs, what would it be?
2. Is there anything we should have asked?
3. Who else should we talk to? (Ask for introductions.)
4. Could you share anonymised examples of the documents we discussed? (Use the checklist in 05.)

3.11 Consent script (summary)

> "We're from TechVaults. We're researching how cooperative societies operate so we can design better tools. There are no right or wrong answers. You can skip any question or stop at any time. With your permission we'll take notes [and record audio]. We will not publish your name or your society's name without written permission. We will not collect member personal data. Any documents you share should be blank or anonymised. Do you agree to continue?"

Record: date, society ID (pseudonym), role, consent given (Y/N), recording consent (Y/N), documents promised.

3.12 Interview snapshot template (complete within 24 hours)

```
Society ID:            Segment (A–H):          State:
Interviewee role:      Interviewers:           Date:
Size / channel / tools (§1.2 axes):

Key story (one concrete episode, step by step):
Opportunities / pain points (quote + impact: H/M/L):
Rules discovered (eligibility, interest, allocation, exit…) + evidence (doc ref):
Workarounds observed:
Surprises / contradictions with prior interviews:
Documents promised / received (ref IDs):
Follow-ups:
```

4. Workflow Checklist

Use this checklist **for each society** to map its current-state workflows. For each item, record:

- **Who** does it (role), **when** (trigger and frequency), **how** (tool or channel), **rule** (the policy, with a source document ref), **approval** (who signs off), **exceptions**, and **pain** (H/M/L).
- Mark each item **Confirmed** (seen in a document or demonstrated), **Described** (told to us only), or **Not applicable**.

Items marked ⚖️ need legal or accounting confirmation before we encode them.

4.1 Membership

- ☐ Who is eligible: employees only, contract staff, retirees, spouses, external members?
- ☐ Member categories (ordinary, associate, honorary, corporate) and their rights
- ☐ Application channel (paper form, online, HR onboarding pack)
- ☐ Required documents (ID, staff ID, passport photo, payslip, NIN/BVN?) and the reason each is required ⚖️ (NDPA minimisation)
- ☐ Employment verification: who confirms, and how
- ☐ Next-of-kin and nominee details: how many, can they change, how is a change witnessed?
- ☐ Entrance fee, initial shares, minimum first contribution
- ☐ Approval: who approves, and is there a waiting period?
- ☐ Member number format and assignment rules
- ☐ Deduction authority form signed for the employer
- ☐ Welcome communication and passbook or card issued
- ☐ Changes: department, salary, grade, name (marriage), bank account, contact details. Who may edit, and is the change approved?
- ☐ Duplicate detection: how a returning or duplicate member is spotted
- ☐ Suspension: grounds, effect on deductions and loans, reactivation
- ☐ Transfer: to another branch, employer or society; what moves with the member
- ☐ Revalidation data: does the society hold what NCSR and CoopID will need? ⚖️

4.2 Savings, shares and contributions

- ☐ List every scheme: share capital, ordinary savings, special or target savings, fixed deposits, thrift, welfare, building fund, Christmas savings, *ajo/esusu*
- ☐ For each scheme: compulsory or voluntary; fixed amount, percentage of salary or member-chosen; minimum and maximum; when it can change
- ☐ Withdrawable or not; notice period; partial withdrawals; penalties
- ☐ Does the scheme earn interest or a return? How is it computed and when credited? ⚖️
- ☐ Share capital: par value, maximum holding, transfer between members, refund on exit ⚖️ (IFRIC 2)
- ☐ Contribution change requests: form, approval, effective payroll month
- ☐ Voluntary lump-sum payments by transfer or cash: receipting, posting
- ☐ Member statements: frequency, format, delivery

4.3 Payroll deductions (monthly cycle)

- [ ] Calendar: cut-off for changes, schedule submission date, payroll run, remittance date, reconciliation deadline
- [ ] Building the deduction schedule: savings, shares, loan repayments, fees, special levies per member
- [ ] File format and submission channel to the employer (Excel template, IPPIS upload, email, portal)
- [ ] Net-pay floor or statutory cap check, and who performs it
- [ ] Remittance: single lump sum or itemised; bank account; proof
- [ ] Returned deduction report: what the employer sends back
- [ ] Reconciliation: expected vs actual per member; handling of **short**, **zero**, **excess**, **duplicate**, **unknown member**, and **late** deductions
- [ ] Allocation order when a deduction is partial (fees, then interest, then principal, then savings?) ⚖️
- [ ] Arrears carried forward, member notified, manual top-up allowed
- [ ] Multiple employers or payroll sources per society
- [ ] Non-payroll members: bank transfer, cash, agent collection; receipts and banking of cash

4.4 Loans

- [ ] Product catalogue: name, purpose, min/max amount, term, rate, **interest method** (flat, reducing balance, upfront deduction, none), fees, insurance
- [ ] Eligibility rules: months of membership, savings multiple (e.g. 2× savings), salary capacity, number of concurrent loans, years to retirement, no arrears
- [ ] Application: form, supporting documents, channel
- [ ] Guarantors: number, eligibility, maximum guarantee exposure, consent capture, release on repayment
- [ ] Use of the borrower's savings as collateral or lien
- [ ] Approval chain: thresholds by amount; conflict-of-interest rules for officers' loans
- [ ] Waiting list and rationing when funds are short
- [ ] Disbursement: method, deductions at source (fees, insurance, upfront interest), offset of an existing loan
- [ ] Repayment schedule generation; start month
- [ ] Repayment allocation; overpayment handling
- [ ] Top-up and refinancing; rescheduling; moratorium
- [ ] Early repayment: interest rebate or not
- [ ] Arrears ageing; reminders; escalation; recovery from guarantors and from savings ⚖️
- [ ] Write-off approval and later recovery ⚖️
- [ ] Loan register and portfolio reports (portfolio at risk, ageing)

4.5 Accounting and financial management

- [ ] Chart of accounts: who designed it, and is it state-prescribed? ⚖️
- [ ] Books kept: cashbook, general ledger, members' ledgers, loan ledgers, fixed-asset register
- [ ] Receipts and payment vouchers: numbering, approval, signatories
- [ ] Journals: who prepares, who approves
- [ ] Bank accounts and reconciliation frequency
- [ ] Interest income recognition; fee income ⚖️
- [ ] Expenses and budgets
- [ ] Investments (fixed deposits, shares, real estate) and trading stock

- [ ] Month-end and year-end close; locking of periods
- [ ] Financial statements produced; framework used ⚖️
- [ ] Corrections: reversal entries vs editing; audit trail
- [ ] Do the member sub-ledgers reconcile to the GL control accounts?

4.6 Dividends and surplus

- [ ] Is a distribution made at all? On what (shares, savings, loan interest paid = patronage)?
- [ ] Statutory reserve fund and education fund appropriations ⚖️ (see 02 F2)
- [ ] Proposal by committee; approval at AGM; ministry or auditor clearance ⚖️
- [ ] Eligibility (membership on a cut-off date, minimum balance, no arrears)
- [ ] Calculation basis (year-end balance, average balance, monthly product)
- [ ] Payment: cash, credit to savings, offset against loans
- [ ] Unclaimed dividends

4.7 Retirement, resignation, transfer, death, expulsion

Map **each event type separately**.

- [ ] Trigger: notice from member, HR or next of kin; required evidence (retirement letter, resignation acceptance, death certificate)
- [ ] Notice periods; stop-deduction instruction to payroll
- [ ] Confirm the final contribution received
- [ ] Outstanding loans, interest to date, arrears
- [ ] Offset order: loans netted against savings and shares? ⚖️
- [ ] The member's guarantees on others' loans: release, substitution or retention
- [ ] Pending transactions and claims (welfare, levies, unposted deductions)
- [ ] Share refund rules (immediate, after AGM, after a notice period) ⚖️
- [ ] Pro-rata dividend or surplus entitlement for the part year
- [ ] Clearance sign-offs (loan officer, treasurer, secretary, chairperson)
- [ ] Death: nominee vs next of kin vs estate; documents; insurance or welfare benefit ⚖️
- [ ] Settlement computation sheet; approval; payment; acknowledgement
- [ ] Shortfall: what happens if loans exceed savings (guarantors, terminal benefits, legal action)?
- [ ] Final statement; closing the account; record retention; rejoining rules
- [ ] Typical elapsed time and complaints

4.8 Governance and audit

- [ ] Officers and committees; terms; elections
- [ ] Approval matrix (who can do what, and value limits)
- [ ] Segregation of duties in practice (does one person prepare and approve?)
- [ ] Meeting minutes and resolutions linked to actions
- [ ] AGM: notice, quorum, voting method, minutes ⚖️
- [ ] Internal audit or supervisory committee
- [ ] External audit: auditor appointment, timing, fee, filing with the state ⚖️
- [ ] Annual returns and other filings to the State Department ⚖️
- [ ] Dispute and complaint handling

- [ ] Record retention and archiving ⚖️

4.9 Technology and data

- [ ] Every tool used, and which workflow lives where (paper, Excel, WhatsApp, accounting package, cooperative software, bank portal)
- [ ] Where the "source of truth" for balances is, and who holds it
- [ ] Backups: do they exist, and who has copies?
- [ ] Who has access to member data; password sharing
- [ ] Devices used by officers and members; internet reliability; electricity
- [ ] Data volume: members, transactions per month, years of history to migrate

5. Sample Document Collection Template

5.1 Rules for handling documents

1. **Ask for blank forms first.** Most of what we need is in the structure of a form, not its contents.
2. **Where filled examples are necessary** (deduction files, settlement sheets, loan schedules), the society anonymises them **before** handing them over. Replace names with `MEMBER-001`, remove staff IDs, NIN, BVN, account numbers, phone numbers and addresses, and keep amounts and dates. Offer to sit with them while they do it. We never do it ourselves on their machine.
3. **Never accept** documents containing NIN, BVN, bank account numbers, health information or ID images. If received by mistake, delete them, record the deletion, and tell the society. ⚖️ NDPA.
4. Store documents in an access-restricted research folder, not on personal devices or in WhatsApp. Retention: until the end of Stage 3, then delete unless the society consents to longer retention.
5. Every document gets an ID: `DOC-<SocietyID>-<NN>`.
6. Get a signed **document sharing note**: what was shared, the permitted use (research and synthetic test data only), and the retention period.

5.2 Request checklist

Code	Document	Blank or anonymised	Priority	Why we need it
D01	Registration certificate and bye-laws	Public or blank	**Must**	Source of policy rules
D02	Approved policies (loan policy, savings policy, exit policy)	As-is	**Must**	Configurable rules
D03	Membership application form	Blank	**Must**	Member data model
D04	Member register (structure only, or 5 anonymised rows)	Anonymised	Should	Fields and data quality
D05	Deduction authority form	Blank	Should	Consent and employer interface
D06	Monthly payroll deduction schedule sent to employer	Anonymised	**Must**	File format, schedule logic
D07	Employer's returned deduction report and remittance advice	Anonymised	**Must**	Reconciliation logic
D08	Contribution change form	Blank	Could	Workflow
D09	Loan application form	Blank	**Must**	Loan data and eligibility
D10	Guarantor form	Blank	**Must**	Guarantor rules
D11	Loan approval minutes or register	Anonymised	Should	Approval chain
D12	Loan repayment schedule for 2–3 real loans	Anonymised	**Must**	Interest method validation
D13	Loan ageing or arrears report	Anonymised	Should	Recovery workflow

Code	Document	Blank or anonymised	Priority	Why we need it
D14	Receipt, payment voucher, cashbook page	Blank, plus one anonymised	Should	Accounting controls
D15	Bank reconciliation statement	Anonymised	Should	Reconciliation design
D16	Trial balance and chart of accounts	As-is (society-level only)	**Must**	Ledger design
D17	Audited financial statements (last 2 years)	As-is (society-level)	**Must**	Reporting outputs, accounting treatment
D18	Auditor's management letter	As-is	Should	Control weaknesses
D19	Dividend or surplus computation	Anonymised	Should	Distribution logic
D20	Retirement or exit settlement sheet (2–3 cases, including one with a loan)	Anonymised	**Must**	Settlement logic
D21	Death claim file checklist	Blank	Could	Next-of-kin process
D22	Member statement	Anonymised	Should	Member UX
D23	Annual returns or filings to the State Department	Blank or as-is	Should	Compliance outputs
D24	NCSR or revalidation forms received	As-is	Should	Regulatory data fields
D25	Screenshots of current software or spreadsheets	Anonymised	Could	Current tooling

5.3 Document intake log (one row per document)

Field	Example
Document ID	DOC-UNI02-12
Society ID / segment / state	UNI02 / A / Oyo
D-code	D12
Title as named by society	"Loan Amortisation – Regular Loan"
Format	XLSX
Received from (role) and date	Treasurer, 2026-11-14
Anonymisation verified by	Researcher initials
Contains prohibited data?	No
Rules extracted (refs to requirements matrix)	R-LN-014, R-LN-015
Retention end date	End of Stage 3
Notes	Flat interest at 1%/month on original principal; upfront 2% form fee

5.4 Rule-extraction template

For each rule found in a document or interview:

```
Rule ID:        R-<AREA>-<NNN>        (AREA: MEM, SAV, PAY, LN, ACC, DIV, EXIT, GOV)
Statement:      e.g. "Regular loan max = 2 × total ordinary savings, cap ₦5,000,000"
Society / source: UNI02 / DOC-UNI02-02 §4.1   or   STAKEHOLDER: UNI02 TREASURER
Evidence level: Document | Demonstrated | Described
Varies across societies? Values seen: UNI02=2×, COR01=3×, MDA01=2× (cap ₦3m)
Candidate classification: Core | Configurable policy | Optional module | Integration | Needs
legal/accounting validation
```

6. Competitor Analysis Framework

Vendor marketing is REPORTED at best. A capability counts as verified only after a demo, a trial account, or a conversation with a current customer.

6.1 Initial long-list (found 17 Oct 2026)

Product	Positioning (vendor's own words, summarised)	Source
Hyves	Staff cooperatives: payroll deduction, loans, contributions, wallets, bookkeeping	hyves.ng
Coopra	Members, staff deductions, savings, thrift, loans, repayments, reports; employee-cooperative page	getcoopra.com
TrustCoop	Staff multipurpose cooperatives; offline mode with sync; tiered pricing from about ₦30,000/month for 100 members	trustcoopapp.com
Coopify	Cooperatives and agencies; free tier for small groups	coopify.africa
Coopco	Cooperative management and accounting; from about ₦15,000/month	coopco.com.ng
ProbityCoop	Dues reconciliation; cloud, on-premise or SaaS	probitysoftware.com
Orbital	Cooperative accounting, loans, payroll module	orbital.com.ng
FlexiCoop	Cooperative and lending management	flexicoop.com
PRIMAX (and other desktop packages)	Loans, payroll, multi-branch	Nairaland thread
Excel + WhatsApp	The real default competitor (hypothesis)	Interviews
Generic accounting tools (QuickBooks, Sage, Zoho Books) with spreadsheets	Common in more formal societies (hypothesis)	Interviews
Apache Fineract / Mifos X	Open-source core-banking for MFIs and credit unions; a reference model, not a direct competitor	fineract.apache.org

Add to this list from interviews: ask every society what they have tried and dropped.

6.2 Evaluation scorecard

Score each criterion 0–3: 0 = absent, 1 = claimed only, 2 = seen in a demo, 3 = verified in use by a customer. Record the evidence.

Area	Criteria
Member lifecycle	Onboarding, custom fields, next of kin, status changes, transfer, exit
Savings	Multiple schemes, share capital vs savings separation, withdrawal rules, interest on savings

Area	Criteria
Payroll	Schedule generation, employer-specific templates, IPPIS or state formats, import of returns, exception handling for short, excess and unknown deductions
Loans	Product configurability, interest methods, eligibility engine, guarantor exposure tracking, approval chains, top-ups, arrears ageing
Exit and settlement	Settlement calculator, offset rules, guarantor release, death claims, final statement
Accounting	True double entry, GL with member sub-ledgers reconciled, bank reconciliation, period close, financial statements, reversals not edits
Dividends	Reserve appropriation, configurable basis, AGM approval step
Governance and audit	Role-based permissions, maker-checker, immutable audit log, auditor read-only access
Multi-society	Multi-tenant, branding, per-tenant configuration, federation or apex reporting
Configurability	Admin-configurable rules without vendor code changes; versioned policies
Security and privacy	MFA, encryption, NDPA posture, data residency, breach process, vendor support access controls
Reporting and export	Standard reports, custom reports, CSV and Excel export, API
Migration and onboarding	Import tools, opening-balance validation, time to go live
Member mobile experience	Mobile web or app, USSD, statements, loan application, notifications
Offline and low-bandwidth	Behaviour on poor connections
Regulatory readiness	CVN and CoopID fields, state filing outputs
Commercials	Pricing model (per member, flat, setup fee), contract terms, data exit
Support	Channels, hours, training, local presence

6.3 Method

1. **Desk scan** (week 2): fill the "claimed" column from websites, brochures and app-store listings.
2. **Demos and trials** (weeks 3–6): request demos for the top 5. Use a **standard scripted scenario** so results are comparable: - Onboard a member with 2 schemes; run a month's deductions with 1 short and 1 unknown deduction; issue a reducing-balance loan with 2 guarantors; miss 2 repayments; retire the member with an outstanding loan; produce the trial balance and member statement.
3. **Customer voice** (during interviews): ask societies that use each product what they like, what they work around, and why they would leave.
4. **Synthesis:** a positioning map (configurability vs accounting depth), a gap list backed by evidence, and an April Dunford-style positioning hypothesis: competitive alternatives, unique attributes, value, target customers, market category.

Rule: a gap is a gap only if a cooperative suffers from it (interview evidence), not just because a competitor lacks a feature.

6.4 Desk scan results (week 0, 17 Oct 2026)

Website claims only. Every non-zero score here is **1 = claimed**, nothing higher. "—" means the site does not mention it. Coopra, FlexiCoop and TrustCoop pages could not be read (no text content or HTTP 404) and need a manual visit.

Area	Hyves	Coopify	Coopco	ProbityCoop
Member lifecycle (onboard → exit)	1 (profiles, history; no exit)	1 (enrolment, updates; no exit)	1 (membership mgmt)	1 (enrolment, member portal)
Savings / share capital separation	1 (savings; no shares)	1 (savings; no shares)	1 (savings, shares, deposits)	1 (savings, contributions)
Payroll deduction + reconciliation	1 (deduction; no reconciliation)	—	—	1 ("automated dues reconciliation")
Loans: interest methods	—	—	1 ("control loans & rates")	—
Loans: guarantors	—	—	1 (guarantor verification)	1 (store context only)
Loans: approval chains	—	1 (admin approval)	—	1 (online approval)
Exit / retirement settlement	—	—	—	—
Double-entry GL, bank reconciliation	1 ("automated ledgers"; no bank rec)	—	1 (cash book, ledgers, trial balance, balance sheet)	—
Dividends / surplus with reserve rule	—	—	—	—
Roles + audit trail	1 (audit trail)	1 (roles, limits)	1 (audit trail on top tier)	1 (Exco/member roles)
Multi-society / multi-branch	1 (multi-branch)	1 (multi-location)	—	—
Member mobile app / USSD	1 (iOS and Android; no USSD)	1 (iOS and Android; no USSD)	1 (custom app)	SMS only
Security / NDPA	1 (generic encryption claim)	1 (generic)	—	1 (generic)
Migration	1 (data import)	—	—	—
Pricing	Free to start	Free / per-member SaaS / self-host	~~₦~~15k / ~~₦~~30k / ~~₦~~90k per month	SaaS / enterprise / self-host; 30-day trial
Named customers	Vanni Intl Security MPCS; APCAA Greenfields MPCS	None ("500+ cooperatives")	None	SterlingBank Staff Coop; Irorun MPCS; Fidelity Bank Credit MPS

Early signals, all HYPOTHESIS **until demos:** 1. **No vendor publicly claims exit or retirement settlement, or dividend computation with the statutory reserve (Act s.34).** Both may be real gaps. Only interviews can show whether societies suffer from them (A20). 2. **Nobody mentions NDPA compliance or the NCSR (CVN, CoopID).** Regulatory readiness may be a differentiator. 3. **Corporate staff cooperatives (banks) are a proven buyer segment.** ProbityCoop names bank staff cooperatives as customers. This supports the pilot recommendation (10). 4. **Prices cluster at ₦15k–₦90k per month** for SMB tiers. That is a useful price anchor for the willingness-to-pay test (A3).

Next: request demos from Hyves, Coopco and ProbityCoop, plus whichever of Coopra, TrustCoop and FlexiCoop respond, using the standard scenario in §6.3.

7. Assumptions Register

None of these statements may be treated as fact. Each has a test, and an owner records its status: **Untested**, **Supported**, **Refuted** or **Partly true** (with the conditions).

#	Assumption	Why it is risky if wrong	How we test it	Status
A1	Workplace cooperatives (segments A–D) share one operating core	We could build the wrong core, or four products	Compare rule matrices across ≥6 workplace societies	Untested
A2	Payroll deduction is the main collection channel for workplace societies	The reconciliation module could be wrongly prioritised	Interview count; deduction file samples	Untested
A3	Societies are willing to pay for software, and someone holds a budget	No business	Ask for current spend and who decides; test price anchors	Untested
A4	Officers would trust a cloud system with member financial data	Adoption fails	Interviews; ask about previous software experiences	Untested
A5	Members want self-service on their phones	Member app over-invested	Member interviews; device survey	Untested
A6	Interest is usually flat-rate on the original principal	Wrong calculation engine	Collect D12 schedules from ≥6 societies	Untested
A7	Guarantors are always other members	Guarantor model wrong	D10 forms	Untested
A8	A member's savings can be offset against their loan at exit	Settlement errors, legal disputes	Bye-laws + legal review	**Partly supported:** Act s.16 gives the society a charge and set-off over a member's shares, deposits and dividends; order and scope still need bye-laws + counsel
A9	Every society pays dividends	Unnecessary module, wrong accounting	Interviews + D17, D19	Untested
A10	The 25% reserve-fund rule applies to all societies	Wrong distribution logic	Legal review of Act + state laws	**Supported for the federal Act** (s.34(2), with Director exemption possible); state laws not yet checked
A11	State laws largely mirror the federal Act	Per-state rules missed	Obtain 3 state laws; counsel review	Untested. Note: the minimum-membership figure differs (Act s.3: 10; guides: 20 for savings and credit)
A12	Cooperatives taking member savings need no CBN licence, and our platform needs none	Regulatory exposure	Counsel opinion	Untested

#	Assumption	Why it is risky if wrong	How we test it	Status
A13	Employers will accept a standard deduction file format from us	Integration effort underestimated	Payroll officer interviews; D06, D07 samples	Untested
A14	IPPIS or state platforms allow cooperatives to submit or receive deduction data digitally	Government segment blocked	Interview MDA payroll staff; official guidance	Untested
A15	Societies have clean opening balances that can be migrated	Pilot stalls on data cleanup	Data-quality review of anonymised registers	Untested
A16	Officers have reliable internet and laptops	Offline needs ignored	Technology questions in §4.9	Untested
A17	NCSR will expose an API or data standard we can integrate with	Wasted integration work	Ask FMAFS, Seamfix, CFN	Untested
A18	The NCSR revalidation creates demand for digital records	Wrong go-to-market timing	Ask societies whether revalidation pushes them to digitise	Untested
A19	One person per society can authorise configuration changes	Governance mismatch	Governance interviews	Untested
A20	Retirement and exit settlement is a major pain point	Mis-prioritised roadmap	Pain ranking from officers and retired members	Untested
A21	Societies will let TechVaults host data, not insist on on-premise	Architecture mismatch	Interviews; ask about IT policy at parent institutions	Untested
A22	Member counts are as large as press reports suggest (30k vs 370k societies conflict)	Market sizing wrong	Request data from FMAFS and CFN	Untested
A23	A rules engine flexible enough for all segments can stay simple to configure	Product becomes unusable	Prototype configuration UI tested with officers	Untested

8. Discovery Timeline, Deliverables and Success Criteria

RECOMMENDATION A 12-week discovery, starting the week of **2 November 2026** and ending **late January 2027**. That is before the NCSR revalidation window is expected to close (about February 2027, if six months from 12 August 2026 holds).

Note: the December and January holiday period will slow access to societies. Book interviews for weeks 3–7 early.

8.1 Plan

Weeks	Phase	Activities	Deliverables
1–2	**Set-up and desk research**	Engage counsel and a cooperative auditor; obtain the Act, 3 state laws and the GAID text; competitor desk scan; recruit societies through CFN, alumni, employers and personal networks; finalise consent forms	Verified legal-source pack, recruitment list (target 15 to get 8–12), competitor long-list scores, ethics and consent pack
3–7	**Field interviews**	8–12 societies × 3–5 roles (about 35–50 interviews); document collection; observation of a month-end close or deduction reconciliation where possible	Interview snapshots, document intake log, rule-extraction entries
5–8	**Workflow mapping**	Current-state maps per segment for membership, the deduction cycle, loans, accounting close, exit; competitor demos with the standard scenario	Swimlane workflow diagrams, pain-point heatmap, competitor scorecard
8–10	**Synthesis**	Requirements matrix; classification into core, configurable, optional, integration, needs-validation; expert reviews (legal, accounting, DPO)	Draft Discovery Report, requirements matrix v1, expert review memos
10–11	**Concept testing**	Clickable prototypes of 3 critical flows (deduction reconciliation, loan application and approval, exit settlement), tested with 5–8 officers and members	Usability findings, revised flows
12	**Decision**	Readout to TechVaults leadership; select pilot; agree MVP scope boundaries	**Cooperative Sector Discovery Report**, pilot recommendation, Stage 2 plan

8.2 Team and roles (minimum)

- Research lead (product): owns the plan, interviews, synthesis
- Business analyst: workflows, rule extraction, requirements matrix
- Designer: observation, prototypes, usability tests
- Engineer (part-time): data and integration feasibility, file formats, ledger questions
- Advisers on retainer: cooperative lawyer, cooperative auditor (ICAN), DPO or compliance organisation

8.3 Success criteria (discovery is "done" when)

1. **Coverage:** at least 8 societies across at least 3 segments and at least 2 states. At least 3 roles per society on average, including at least 6 treasurers or accountants and at least 4 exited or retired members.
2. **Evidence:** at least 5 societies provided the D-codes marked **Must** in 05, including at least 10 anonymised loan schedules and at least 6 exit settlement sheets.
3. **Rules:** the requirements matrix shows, for each lifecycle area, the observed **range of variation** across societies, with evidence references.
4. **Assumptions:** every row in 07 has moved from Untested to a status backed by evidence.
5. **Expert sign-off:** written memos from legal, accounting and data-protection advisers answering 02 §2.6, or explicitly deferring each question.
6. **Pain ranking:** the top 10 operational problems ranked by frequency × severity × willingness to pay, each backed by quotes.
7. **Pilot:** at least one society has agreed in writing to pilot, with named champions and data-sharing terms.
8. **Prototype signal:** in usability tests, at least 4 of 5 officers complete the deduction-reconciliation prototype task without help.

9. Turning Findings into a Configurable Product Requirements Document

9.1 Pipeline

```
Interviews + documents
      │  (rule-extraction template, 05 §5.4)
      ▼
Rule inventory  ──►  Variation matrix (rule × society)
      │
      ▼
Classification (5 buckets)  ──►  Expert review on ⚖ items
      │
      ▼
Policy model (what is configurable, within which guardrails)
      │
      ▼
PRD: use cases, workflows, permissions, acceptance tests, MVP cut
```

9.2 The variation matrix

Rows are rules (e.g. "loan eligibility: minimum months of membership"). Columns are societies. Cells hold the observed value and an evidence reference. This makes the classification evidence-based:

Pattern observed	Classification
Same rule everywhere, or mandated by law	**Core**: hard-coded behaviour, not configurable
Same rule *shape*, different values (6 vs 3 months; 2× vs 3× savings)	**Configurable policy**: a parameter
Different rule shapes (flat vs reducing interest; different allocation orders)	**Configurable policy**: choose from a catalogue of *strategies* we have validated, not a free-form formula
Used by some segments only (*ajo*, commodity sales, welfare)	**Optional module**
Depends on an external party's format (payroll, bank)	**Integration**: templates and adapters
Accounting or legal treatment unclear	**Needs validation**: blocked until an expert memo exists

RECOMMENDATION Prefer *parameters* and *named strategies* over a general-purpose rules or formula language. A strategy catalogue can be tested, explained to auditors and shown in plain language in the UI. Arbitrary formulas cannot. Add a new strategy only when a real society needs it and an accountant has verified it.

9.3 Guardrails (non-configurable)

These follow from the master prompt §§7–9. They are confirmed or extended by expert review.

- Double-entry integrity; posted entries are corrected only by reversal
- Statutory requirements (e.g. minimum reserve appropriation, if confirmed by counsel)
- Segregation of duties for prepare, approve, post, reverse and audit
- Audit logging of every configuration change, with approver
- **Policy versioning:** each transaction stores the policy version that produced it; a change applies from an effective date and is never retroactive without an explicit, approved recalculation
- Tenant isolation; TechVaults support access only with time-boxed, logged authorisation

9.4 PRD structure (Stage 2 output)

1. Problem statements and target segment for the MVP (evidence-linked)
2. Personas and jobs-to-be-done (officer, treasurer, loan officer, payroll officer, auditor, member)
3. Role and permission matrix
4. Use cases per lifecycle area, each with: - Business rules (by Rule ID) - Workflow diagram (to-be) - Configuration parameters and guardrails - **Acceptance criteria** written as examples (Given / When / Then), with **worked monetary examples verified by the accountant** - Edge cases (short deduction, concurrent approval, duplicate upload, mid-loan policy change, exit with guarantees outstanding) - Reports affected and how they reconcile
5. Integration specs (payroll file templates, bank statement import)
6. Non-functional requirements: security, privacy (DPIA outcomes), availability, backup and restore targets (RPO/RTO), performance at expected volumes, accessibility (WCAG 2.2 AA), low-bandwidth behaviour
7. Data migration requirements and opening-balance validation
8. MVP scope, explicit non-goals, and roadmap by stage
9. Open questions and risks

9.5 Traceability

Every requirement links back to Rule IDs, evidence (document or interview), and acceptance tests. A requirement with no evidence link is labelled `HYPOTHESIS` and cannot enter MVP scope without a validation plan.

10. Choosing the First Pilot Cooperative

10.1 Recommendation

RECOMMENDATION Pilot with **one mid-sized workplace (payroll-deduction) cooperative**: about **300–2,000 members**, a **single employer and payroll source**, and **savings plus 2–4 loan products**. A private-company or university staff cooperative is most likely to fit. Confirm the choice against discovery evidence in week 12.

Why: - Payroll-deduction societies have the most structured, repeatable monthly cycle. That makes balances easy to verify against existing records. *(Hypothesis A2.)* - A single payroll source avoids IPPIS or multi-employer complexity in the first release. It still exercises reconciliation, loans, guarantors and exit. - Mid-size: large enough to feel real pain and pay, small enough to reconcile every member by hand during the pilot. - University or corporate societies are more likely to have an accountant, an external auditor and decent connectivity. These are the conditions for validating ledger correctness.

Avoid as the first pilot: a federal MDA society on IPPIS (integration and bureaucracy risk), a multi-branch multipurpose society (scope), and community or *ajo* groups (different collection model). Study them in discovery; serve them later.

10.2 Selection scorecard (score 1–5, weighted)

Criterion	Weight	What "5" looks like
Fit with the likely MVP segment	20%	Payroll deductions, savings + loans, single employer
Data quality and accessibility	15%	Electronic records for ≥2 years, reconciled audited accounts, willing to anonymise
Committed champions	15%	Treasurer **and** chairperson actively want change; time allocated
Governance approval	10%	Committee resolution to pilot; clear signatory for data-sharing agreement
Representative complexity	10%	≥2 loan products, guarantors, some exits per year, dividends
Auditor cooperation	10%	External auditor willing to review pilot outputs
Commercial potential	10%	Willing to convert to paid; referral value to similar societies
Logistics	5%	Reachable for on-site visits; reliable internet
Regulatory standing	5%	Registered and in good standing; revalidation (CVN) in progress

Disqualifiers: an ongoing fraud investigation or major dispute; no audited accounts in the last 2 years; refusal to run parallel operations.

10.3 Pilot terms to agree

- Parallel run: the society keeps its current system for ≥3 monthly cycles, and balances are reconciled each month

- Data processing agreement (NDPA-compliant), with controller and processor roles agreed
- Use of anonymised or synthetic data until security review sign-off
- Success metrics: 100% of member balances match after migration (or every difference explained); month-end close time reduced; zero unexplained ledger differences; officer satisfaction
- Exit plan: full data export in open formats if the pilot ends

11. UX and UI Direction (Provisional)

Status: RECOMMENDATION, provisional. The master prompt (§10) says not to fix the technology stack before requirements are documented. This document records the **design direction** and a **preferred UI toolkit** so prototypes in weeks 10–11 can be built with components we expect to keep. Confirm in Stage 4.

11.1 What "premium" means for this product

A cooperative platform feels expensive when it feels **trustworthy, calm and exact**, not when it is decorated. The Linear and Stripe reference points:

- **Numbers are the hero.** Tabular (monospaced) figures, right-aligned amounts, consistent ₦ formatting, negative values that cannot be misread, and every balance one click from its breakdown ("How was this calculated?").
- **Restraint.** A neutral palette, one brand accent, generous spacing on member screens and dense layouts on accountant screens. Colour carries status (posted, pending, reversed, overdue), never decoration.
- **Speed you can feel.** Instant navigation, optimistic UI only where it is safe (never for posting money), a command palette (⌘K) for officers, and keyboard-first workflows for treasurers.
- **Every state designed.** Loading skeletons, empty states that teach, errors that say what to do next, permission-denied states that name who can help, and reconciliation warnings that cannot be ignored.
- **Explainability.** Approval chains, policy versions and audit history are visible in context: "Approved by Treasurer, 12 Nov · Policy v3".
- **Inclusive by default.** WCAG 2.2 AA, 16px+ body text on mobile, large touch targets, plain English (with Yoruba, Hausa and Igbo localisation if discovery supports it), and graceful behaviour on slow 3G and older Android phones.

11.2 Preferred UI toolkit (assuming React + TypeScript, to be confirmed in Stage 4)

Need	Choice	Why
Component foundation	**shadcn/ui** on **Radix UI** primitives	We own the source (copied into the repo, not a black-box dependency), accessible primitives, full design control, the industry-standard look Linear and Vercel users expect
Styling	**Tailwind CSS** with design tokens as CSS variables	Consistent spacing and type scale; per-tenant branding by swapping tokens; light and dark themes
Data tables (ledgers, registers, schedules)	**TanStack Table** + **TanStack Virtual**	Headless, so it fits shadcn styling; sorting, filtering, grouping, column pinning; virtualises 10k+ rows

Need	Choice	Why
Spreadsheet-grade grids (reconciliation, bulk deduction edits)	**AG Grid** (evaluate Community vs Enterprise)	Excel-like keyboard editing, range selection, clipboard paste. Only where treasurers truly need spreadsheet behaviour.
Forms and validation	**React Hook Form** + **Zod**	Long multi-step forms (membership, loans) with schema validation shared with the API
Server state	**TanStack Query**	Caching, retries, background refresh, good offline-tolerant behaviour
Charts	**Recharts** via shadcn chart components (or **Tremor** for dashboard blocks)	Consistent with the design system; enough for portfolio, arrears and contribution trends
Command palette	**cmdk**	⌘K to find a member, loan or report instantly
Notifications	**Sonner**	Quiet, stackable toasts
Date inputs	**react-day-picker** (shadcn Calendar)	Accessible; supports payroll-month pickers
Icons	**Lucide**	Consistent stroke icons, tree-shakeable
Motion	**Motion** (formerly Framer Motion), used sparingly	Micro-interactions (approve, post, reconcile) that confirm state changes; respects reduced-motion
Typography	**Inter** or **Geist** with `font-variant-numeric: tabular-nums` for money	Legible at small sizes; aligned figures
Money in the UI	`Intl.NumberFormat('en-NG', { style: 'currency', currency: 'NGN' })` for display; a decimal library (e.g. **decimal.js** or **dinero.js**) for any client-side arithmetic	Never use JavaScript floats for money; authoritative calculations stay on the server
Component workshop	**Storybook** with accessibility add-on + visual regression	Every component documented in all states before it is used in a screen

Deliberately not chosen: heavy all-in-one kits (MUI, Ant Design) as the base, because their look is hard to make distinctive and they carry large bundles. Ant Design's table and form patterns are still worth studying for enterprise data entry.

11.3 Signature custom components to design

Discovery should validate these. They are where TechVaults can be visibly better than competitors.

1. **Balance card with breakdown**: share capital, savings by scheme, loans, guarantees given, net position; each line expandable to its transactions.
2. **Deduction reconciliation workbench**: expected vs received per member, with exception buckets (short, zero, excess, unknown, duplicate), bulk resolution and a running variance that must reach zero before close.

3. **Loan eligibility explainer**: a checklist-style view showing each rule, the member's value and pass or fail, so officers and members understand a decision.
4. **Guarantor exposure view**: who guarantees whom, how much, and what is at risk.
5. **Approval timeline**: maker-checker steps with actor, time, comment and policy version.
6. **Exit settlement calculator**: a step-by-step statement (savings + shares + dividends – loans – pending items = payable or shortfall), with the clearances checklist.
7. **Audit trail drawer**: for any record, who changed what and when, before and after; read-only for auditors.
8. **Policy editor with preview**: change a parameter, see the effect on sample members before submitting for approval, and see the effective date.

11.4 Two experiences, one design system

- **Officer console (desktop-first, responsive):** dense tables, keyboard shortcuts, side panels, bulk actions, export everywhere.
- **Member app (mobile-first PWA; USSD and SMS considered in discovery):** balances, statements, loan application and status, guarantor requests to accept or decline, and notifications. Performance budget: interactive in under 3s on a mid-range Android over 3G (to be validated).

11.5 How design runs during discovery

- Weeks 3–7: designers attend interviews and collect real screenshots and spreadsheets (anonymised). Real layouts beat assumptions.
- Weeks 8–9: low-fidelity flows for the 3 critical journeys.
- Weeks 10–11: high-fidelity clickable prototypes built with the toolkit above (Figma, or coded prototypes using shadcn/ui), tested with officers and members.
- Output: a design-principles page, a token set (colour, type, spacing, radius, elevation), and a component inventory for Stage 4.

Adviser Terms of Reference

Three short engagements for weeks 1–10 of discovery (decision D-004). Send each brief with a request for a fee quote and a note of any conflicts of interest (for example, an existing relationship with a cooperative software vendor).

Common terms for all three: - **Client:** TechVaults - **Purpose:** advise on the design of a configurable cooperative management platform for Nigerian cooperative societies. The engagement is advisory, not a formal opinion for third parties, unless agreed separately. - **Confidentiality:** mutual NDA. No member personal data will be shared with the adviser. - **Deliverable format:** a written memo that answers each numbered question. For each answer, mark it as a firm position or as uncertain, and list what would resolve the uncertainty. - **Timeline:** kick-off in week 1 or 2, interim call in week 5, memo by **week 9**, review session in week 10.

A. Cooperative and financial-services lawyer

Profile: Nigerian legal practitioner with cooperative-law and financial-regulation experience. Familiarity with state cooperative departments is preferred.

Questions (from 02 §2.6, items 1–7, expanded): 1. How do the Nigerian Co-operative Societies Act (Cap N98) and state cooperative laws (Lagos, the FCT, and one more state we will name) relate? Which governs a society registered in a state whose employer is a federal body? Please give a jurisdiction matrix covering registration, audit, annual returns, reserve rules, disputes and liquidation. 2. Confirm the current status of the Act review and any repeal-and-re-enactment bill announced by FMAFS in 2026. Confirm the status of the 2019, 2023 and 2024 amendment bills. 3. Does s.34 (25% reserve, 10% education fund, distribution approvals) apply unchanged to state-registered societies? Where does the "20 members for savings and credit" figure come from, given that s.3 says 10? 4. On exit and death: what are the scope and order of the s.16 charge and set-off, nominee vs heir under s.18, and what documentation a society may demand? 5. CBN: does taking savings from members only fall outside CBN licensing? What changes if a society accepts payments online or through a payment service provider? What must TechVaults avoid so that it is not itself treated as conducting regulated activity? 6. Interest: are there any caps, disclosure rules or consumer-protection obligations (FCCPC) on member loans? 7. Are electronic records, e-signatures, electronic guarantor consent and electronic AGM voting legally effective (Evidence Act 2011 s.84, and other relevant law)? 8. What are the minimum record-retention periods for cooperative financial and membership records? 9. What contract structure should TechVaults use with societies: SaaS terms, liability caps, data ownership and exit rights? 10. NCSR: does the revalidation or the registry impose data or reporting obligations that software vendors must support?

B. Cooperative auditor (ICAN-qualified)

Profile: Chartered accountant who audits cooperative societies and has filed with at least one State Department of Cooperatives.

Questions: 1. Which reporting framework applies in practice (IFRS, IFRS for SMEs, a state-prescribed format)? Does the FRC take a position? 2. Members' shares: are they equity or a liability under IFRIC 2? How should withdrawable and non-withdrawable savings be presented? 3. Interest income: how do flat-rate,

reducing-balance and upfront-deducted interest compare in recognition and presentation? 4. What approach do state auditors expect for loan impairment and provisioning? 5. Is there a standard chart of accounts used by state audit departments? Please share a sample. 6. Dividend and patronage-refund computation: what is the basis, the cut-off, and how does the reserve and education-fund appropriation sequence (s.34) work? 7. Which audit trail, period-locking and reversal behaviours would let you rely on system-generated records? 8. Which reports and evidence do you request at every audit? Please share anonymised examples. 9. **Independently verify** a set of worked examples we will prepare (loan schedules, a deduction allocation, an exit settlement, a dividend run). These become acceptance tests.

C. Data protection officer or licensed compliance organisation

Profile: NDPC-licensed Data Protection Compliance Organisation, or a credentialed DPO.

Questions: 1. Controller, processor or joint controller: what are the roles of TechVaults and of each society? 2. Is a DPIA required (financial data, scale, NIN or CoopID)? Which GAID 2025 audit-return tier would TechVaults fall in? 3. What lawful basis applies to member, next-of-kin, nominee and guarantor data? 4. Are there hosting location and cross-border transfer requirements? 5. What breach-notification obligations and timelines apply to us and to the societies? 6. Please review our research consent form and document-sharing note (consent-and-document-sharing.md). 7. What is a minimum privacy-by-design checklist for Stage 4 architecture?

Recruitment Outreach

Target: contact about 15 societies to secure 8–12 interviews (D-003). Channels: personal and alumni networks, employer HR contacts, the Cooperative Federation of Nigeria (CFN) and state federations, and State Directors of Cooperatives.

Tracker columns: Society ID · Segment · State · Contact role · Channel · Date contacted · Response · Screener done · Interview dates · Documents promised.

1. Email or letter to a cooperative society

Subject: Research on cooperative operations: request for a 1-hour conversation

Dear [Chairperson / Secretary / Treasurer],

TechVaults is studying how cooperative societies in Nigeria run their day-to-day operations: membership, monthly contributions, loans, accounts, and member retirement or exit. We want to understand the real work and its challenges **before** designing any software.

We would be grateful for one-hour conversations with two or three of your officers, for example the treasurer, the loan officer and the secretary, at a time and place convenient to you. We are not selling anything at this stage.

- Your society and officers will not be named in anything we publish without written permission.
- We will not ask for members' personal information. If you choose to share forms or reports, blank or anonymised copies are all we need.
- You will receive a summary of our findings and early access to whatever we build.

If you're willing, please reply with a suitable contact and some possible dates between [date] and [date].

Yours sincerely, [Name], [Role], TechVaults [Phone] · [Email]

2. Letter to CFN or a State Director of Cooperatives

Subject: Request for guidance and introductions: cooperative operations research

Dear [Title and Name],

TechVaults is conducting research into the operations of cooperative societies across institution types (universities, government agencies, hospitals, private companies, community and multipurpose societies). The research will inform a cooperative management platform that supports good governance, accurate records and transparent reporting, in line with the ongoing national cooperative digitalisation programme.

We respectfully request: 1. A short meeting to understand the [Federation's / Department's] priorities, including registration, audit and returns requirements, and the current revalidation exercise. 2. Copies of any public guidance, model bye-laws, or audit and return formats. 3. Where appropriate, introductions to societies that might take part in interviews.

We will share an anonymised summary of our findings with the [Federation / Department].

Yours faithfully, [Name], [Role], TechVaults

3. Follow-up (after 7 days without a response)

> Hello [Name], following up on our note about cooperative operations research. A 30-minute call with one officer would already help us a great deal. Would any time next week suit you?

4. Thank-you (within 48 hours of an interview)

> Thank you for your time on [date]. As agreed, [documents] can be shared [how]. Please remove names, staff IDs, NIN/BVN and account numbers first. We'll send our summary of findings by [month].

Society Screener (10 minutes, by phone)

Use this to place each society in the sample (D-003) and to pre-score it as a pilot candidate (10 §10.2).

#	Question	Record
1	What type of institution is the society attached to, if any?	Segment A–H
2	In which state is it registered? Is it registered with the State or the Federal Director?	State / registrar
3	Roughly how many active members?	<100 · 100–299 · 300–2,000 · >2,000
4	How do members mainly pay: payroll deduction, transfer, cash, or a mix? If payroll, which system (IPPIS, state, company HR)?	Channel / payroll system
5	How many employers or payroll sources?	Number
6	Which services: savings, shares, loans (how many products), welfare, trading, investments?	List
7	How are records kept today: paper, Excel, accounting package, cooperative software (which)?	Tools
8	Were the last accounts audited? Which year?	Year / auditor type
9	Have you started NCSR revalidation? Do you have a CVN?	Status
10	Who could speak with us: treasurer, loan officer, secretary, payroll contact, an auditor, a retired member?	Roles available
11	Would you be open, in principle, to piloting new software alongside your current system later?	Yes / Maybe / No

Include if segment quotas are not yet full and at least 2 officer roles are available. **Flag as pilot candidate** if: payroll channel, one payroll source, 300–2,000 members, audited within 2 years, answer to Q11 is Yes or Maybe, and segment A or D.

Research Consent Form and Document-Sharing Note

Draft. To be reviewed by the data protection adviser before first use (adviser ToR C, Q6).

Part 1: Interview consent

Research: Cooperative operations discovery **Organisation:** TechVaults · Contact: [name, email, phone]

1. **Purpose.** We are learning how cooperative societies operate so that we can design better management software.
2. **What participating involves.** An interview of about 45–60 minutes about your role and your society's processes.
3. **Voluntary.** You may skip any question, stop at any time, or withdraw your answers within 30 days by contacting us.
4. **What we record.** Written notes. Audio is recorded **only if you tick the box below**. Recordings are transcribed and deleted within 30 days.
5. **What we will not collect.** Members' names, staff IDs, NIN, BVN, bank account numbers, salary details of named people, or health information.
6. **Confidentiality.** Your society is given a code (e.g. "UNI02"). Your name and your society's name will not appear in any report or publication without your separate written permission.
7. **Storage and retention.** Notes are kept in an access-restricted TechVaults folder until **[end of Stage 3 date]**, then deleted or fully anonymised.
8. **Your rights.** Under the Nigeria Data Protection Act 2023 you may ask to access, correct or delete the information you provided, and you may complain to the Nigeria Data Protection Commission.

	Yes	No
I agree to take part in this interview	☐	☐
I agree to audio recording	☐	☐
I agree to be contacted for follow-up questions	☐	☐
I agree that anonymised quotes may be used in internal TechVaults reports	☐	☐

Name: _____ Role: _____ Society code: _____ Date: _____ Signature: _____
Interviewer: _____

Part 2: Document-sharing note

The society **[name]** agrees to share the documents listed below with TechVaults for the **sole purpose** of cooperative operations research and the creation of **synthetic (fictional) test data**.

Conditions: - The society removes all personal identifiers before sharing: names (replace with `MEMBER-001` etc.), staff IDs, NIN, BVN, account numbers, phone numbers and addresses. Amounts and dates may remain. - If TechVaults receives any personal data by mistake, it will delete it, record the deletion, and inform the

society within 5 working days. - Documents will not be shared outside the TechVaults research team or its bound advisers. - Retention until **[date]**, then deletion, unless the society agrees in writing to extend it.

Doc ID	D-code	Description	Format	Anonymised by (society)

For the society: _____ (name, role, signature, date) For TechVaults: _____ (name, role, signature, date)